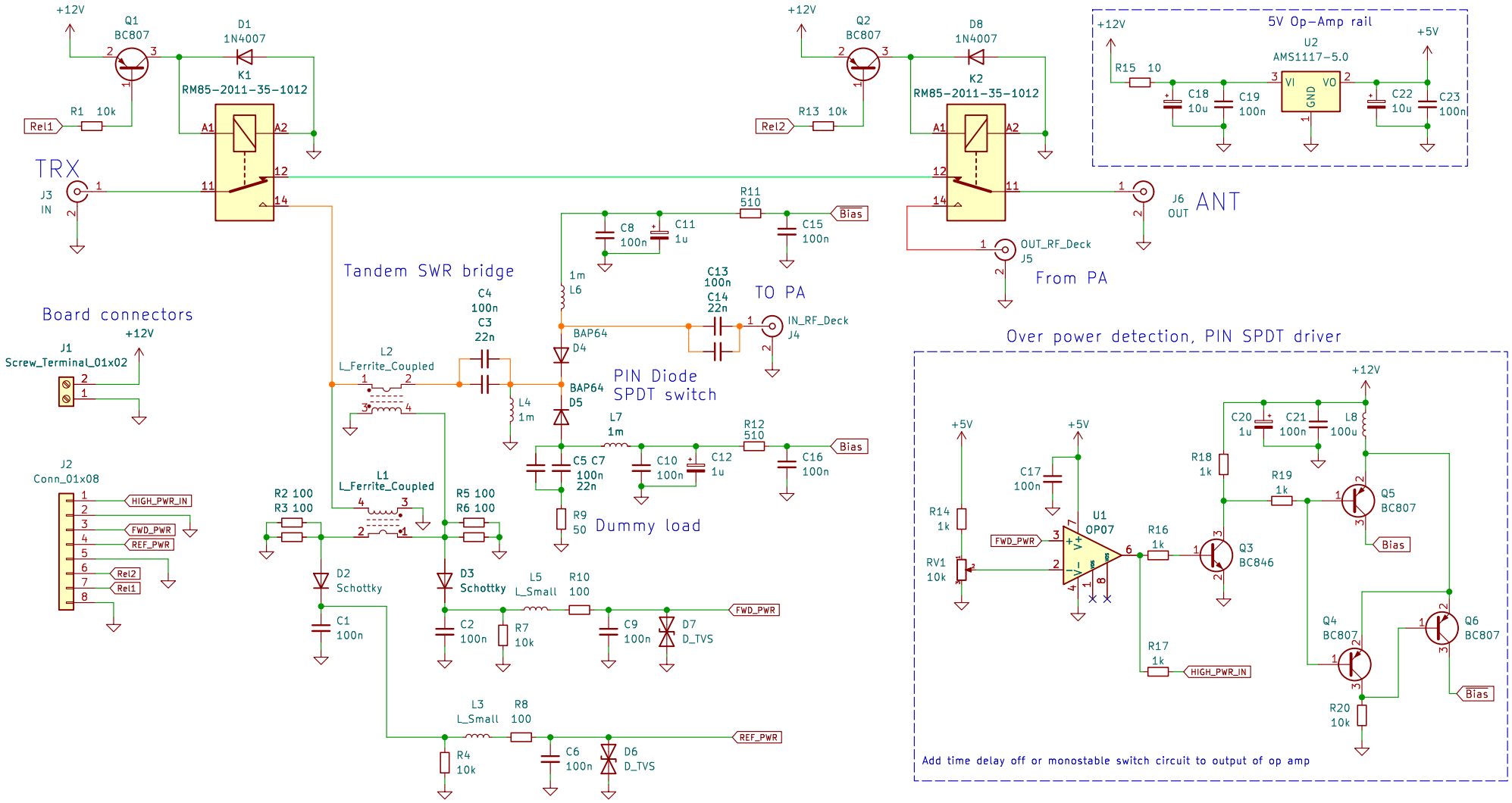
+12V
Q1
BC807
D1
1N4007
K1
RM85-2011-35-1012
R1 10k
Rel1
TRX
J3
IN
Tandem SWR bridge
Board connectors
J1
Screw_Terminal_01x02
J2
Conn_01x08
HIGH_PWR_IN
FWD_PWR
REF_PWR
Rel2
Rel1
L2
L_Ferrite_Coupled
L1
L_Ferrite_Coupled
C4
100n
C3
22n
L4
1m
BAP64
D4
BAP64
D5
L7
1m
C5 C7
100n
22n
C10
100n
C12
1u
R11
510
R12
510
C8
100n
C11
1u
C15
100n
C16
100n
Bias
1m
L6
C13
100n
C14
22n
TO PA
IN_RF_Deck
J4
R9
50
Dummy load
R2 100
R3 100
R5 100
R6 100
D2
Schottky
D3
Schottky
C1
100n
C2
100n
L5
L_Small
R10
100
R7
10k
C9
100n
D7
D_TVS
L3
L_Small
R8
100
R4
10k
C6
100n
D6
D_TVS
Q2
BC807
D8
1N4007
K2
RM85-2011-35-1012
R13 10k
Rel2
J6
OUT
ANT
OUT_RF_Deck
J5
From PA
5V Op-Amp rail
U2
AMS1117-5.0
R15 10
C18
10u
C19
100n
C22
10u
C23
100n
+5V
Over power detection, PIN SPDT driver
U1
OP07
C17
100n
R14
1k
RV1
10k
R16
1k
Q3
BC846
R18
1k
R19
1k
C20
1u
C21
100n
L8
100u
Q5
BC807
Q4
BC807
Q6
BC807
R17
1k
R20
10k
Add time delay off or monostable switch circuit to output of op amp
-Input signal SWR tandem bridge
-Overpower protection
-Antenna to PA deck relay system
SP6GK
Sheet: /
File: TXRX_relay_board.kicad_sch
Title: TX-RX Relay Board
Size: A4
Date: 2026-02-01
Rev: 1.2
KiCad E.D.A. 8.0.9
Id: 1/1